

Credit Default Risk Analysis for Digital Lending

Vitto Technical Assessment

Objective

The objective of this project was to analyze borrower repayment behavior and develop predictive models for identifying credit default risk using the UCI Credit Card Default dataset. The analysis aimed to simulate a real-world FinTech credit-risk workflow involving exploratory analysis, feature engineering, predictive modeling, and business-oriented risk interpretation.

Dataset Overview

- 30,000 credit card clients
- Taiwan credit data (2005)
- 25 variables
- Target: default payment next month
- Includes demographic, financial, and repayment history variables

Key EDA Findings

- Repayment delay variables showed the strongest relationship with future default risk.
- Clients with repeated delinquency patterns exhibited sharply higher default rates.
- Higher credit utilization was associated with increased financial stress.
- Repayment consistency negatively correlated with default probability.
- Behavioral financial variables were substantially more predictive than demographic attributes.

Feature Engineering

- AVG_UTIL_RATE — Average proportion of available credit utilized across six months.
- AVG_PAY_RATIO — Repayment consistency ratio across billing cycles.
- TOTAL_DELAY_MONTHS — Number of months with repayment delays.
- MAX_DELAY — Maximum historical repayment delay observed for a client.

Modeling Approach

Two predictive models were developed:

- Logistic Regression — Interpretable baseline model
- Random Forest — Nonlinear ensemble model capable of capturing feature interactions

The models were evaluated using Precision, Recall, F1 Score, ROC-AUC, and Stratified 5-Fold Cross Validation. Class imbalance was handled using `class_weight='balanced'`.

Results

The Random Forest model achieved stronger predictive performance compared to Logistic Regression by capturing nonlinear repayment behavior patterns and complex feature interactions.

Repayment delay features consistently emerged as the strongest predictors of future default risk.

Explainability & Fairness

SHAP analysis confirmed that repayment delay variables and delinquency-related engineered features were the strongest contributors to predicted default risk.

A preliminary fairness assessment was also conducted by comparing False Positive Rates across demographic groups. While some variation was observed, the analysis remained exploratory and highlighted the importance of fairness monitoring in digital lending systems.

Business Recommendations

1. Implement early delinquency monitoring for borrowers showing repeated repayment delays.
2. Integrate behavioral repayment features into credit scoring systems to improve risk identification accuracy.

Conclusion

The project demonstrates how repayment behavior and financial usage patterns can be leveraged to build effective and interpretable credit-risk prediction systems for digital lending environments.

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